

ABSTRACT

AgriTrace is a comprehensive, full-stack web application designed to address the pressing issue of agricultural waste management in India. The platform provides a digital ecosystem connecting farmers, collection agents, and recycling administrators through a role-based system built on modern web technologies.

The system enables farmers to report and list agricultural waste such as wheat stubble, rice residue, corn stover, and other crop residues. Collection agents can browse available waste listings, claim collection tasks, and manage the pickup-to-delivery workflow. Administrators oversee the entire platform, managing users, monitoring analytics, and tracking the recycling pipeline.

Key features include real-time waste tracking with GPS-based location services, photo documentation for waste verification, an integrated payment workflow for agent-to-farmer transactions via Razorpay, a carbon credit tracking system that quantifies environmental impact, and a gamification-based rewards system to incentivize participation. The platform also integrates AI capabilities through Google Genkit for intelligent waste classification.

Built with Next.js 15.5, TypeScript, Firebase (Firestore, Authentication, Storage), Tailwind CSS, and Radix UI components, AgriTrace delivers a responsive, modern user experience. The application employs Firestore security rules for data protection and supports real-time data synchronization across all user roles.

AgriTrace contributes to environmental sustainability by diverting agricultural waste from open burning—a major cause of air pollution—toward productive recycling channels, while providing economic incentives to all stakeholders in the waste management chain.